Grade 10 – Term 1
Decision-Making Under Uncertainty
Exam Solutions – Version 2A

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Learning objective

Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.

Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

Question	C8	C9
1	3	–
2	3	–
3	–	2
4	–	2
5	–	2
Total	**6**	**6**

Evaluated criteria

C8 Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.

C9 Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

C10 Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

1 Camera Rental Plans [C8-3]

A camera rental studio selects its inventory under uncertain demand. Profits are in thousand USD.

Alternative	High	Normal	Low
$P(S_j)$	0.3	0.5	0.2
Essential (A_1)	110	95	65
Shared (A_2)	150	160	35
Extensive (A_3)	250	52	-35
Modular (A_4)	180	135	-15

Decision statistics (Regret = maximum regret):

Alt.	Max	Min	Regret	EMV
A_1	110	65	140	93.5
A_2	160	35	100	132
A_3	250	-35	108	94
A_4	180	-15	80	118.5

(a) Using the original camera rental payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A subsidy s (integer, $0 \leq s \leq 95$) increases the low-demand payoffs of A_3 and A_4 by s. An alternative is eligible only if all its payoffs are nonnegative. Determine each camera rental plan's eligibility based on the value of s. **[0.5]**

(c) Among eligible camera rental plans, find the smallest allowed subsidy s at which the Maximin recommendation changes from A_1 to A_4 alone. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $0 \leq s \leq 95$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

(a) Initial recommendations [1]

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	A_3	Row maximum: 250
Maximin	A_1	Row minimum: 65
Minimax Regret	A_4	Maximum regret: 80
EMV/Bayes	A_2	EMV: 132

All deciding values are in thousand USD. Maximax pursues the greatest upside; Maximin protects the worst outcome; Minimax Regret limits missed profit; EMV uses the forecast probabilities to maximize expected profit. Regret benchmarks use all four original alternatives, with state bests $(250, 160, 65)$; no eligibility restriction applies in (a).



(b) Subsidy and eligibility [0.5]

The updated payoff table is:

Alternative	High	Normal	Low
$P(S_j)$	0.3	0.5	0.2
Essential (A_1)	110	95	65
Shared (A_2)	150	160	35
Extensive (A_3)	250	52	$-35+s$
Modular (A_4)	180	135	$-15+s$

The updated decision statistics are:

Alt.	Max	Min	Regret	EMV
A_1	110	65	140	93.5
A_2	160	35	100	132
A_3	250	$\min(52, -35+s)$	$\max(108, 100-s)$	$94+0.2s$
A_4	180	$-15+s$	$\max(70, 80-s)$	$118.5+0.2s$

An alternative is eligible only if all its payoffs are nonnegative. For integer subsidies $0 \le s \le 95$:

Alternative	Eligibility condition	Eligible subsidies
A_1	$65 \ge 0$	$0 \le s \le 95$
A_2	$35 \ge 0$	$0 \le s \le 95$
A_3	$-35+s \ge 0$	$35 \le s \le 95$
A_4	$-15+s \ge 0$	$15 \le s \le 95$

Thus, the eligible sets are:

$$\begin{aligned} 0 \le s \le 14: &\quad \{A_1, A_2\}, \\ 15 \le s \le 34: &\quad \{A_4, A_1, A_2\}, \\ 35 \le s \le 95: &\quad \{A_3, A_4, A_1, A_2\}. \end{aligned}$$

(c) Maximin change under subsidy s [1.5]

The minimum payoffs are

$$m_1 = 65, \quad m_2 = 35, \quad m_3(s) = \min(52, -35 + s), \quad m_4(s) = -15 + s.$$

Since $m_3(s) \le 52 < 65 = m_1$ and $m_2 = 35 < 65$ throughout the allowed range, only A_4 can challenge A_1.

For A_4 to beat A_1:

$$-15 + s > 65 \quad \Longleftrightarrow \quad s > 80.$$

Thus, A_1 is the sole Maximin choice for $0 \le s \le 79$; at $s = 80$, A_4 and A_1 tie with minimum payoff 65; and A_4 is the sole Maximin choice for $81 \le s \le 95$. The tie at $s = 80$ first changes the set of recommended plans; a strict change to A_4 alone requires $s > 80$. The integer domain contains a smallest value satisfying this condition. Therefore, the smallest allowed integer subsidy giving a strict change is

$$\boxed{s = 81}.$$

At $s = 81$, all four alternatives are eligible and their minimum payoffs are $65, 35, 46, 66$, respectively. Hence A_4 wins uniquely with a guaranteed profit of 66 thousand USD.

2 Outdoor Cinema Refreshments [C8-3]

A cinema manager selects a refreshment kiosk under uncertain audience size. Profits are in hundred USD.

Alternative	High	Low
$P(S_j)$	0.35	0.65
Portable (C_1)	102	68
Full-service (C_2)	162	24
Deluxe (C_3)	190	-12
Fixed (C_4)	148	52

Decision statistics (Regret = maximum regret):

Alternative	Max	Min	Regret	EMV
C_1	102	68	88	79.9
C_2	162	24	44	72.3
C_3	190	-12	80	58.7
C_4	148	52	42	85.6

(a) Using the original refreshment kiosk payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A sponsor replaces C_1's large-audience payoff of 102 with g (integer, $102 \leq g \leq 170$). An alternative is eligible only if all its payoffs are at least 44. Determine each refreshment kiosk's eligibility based on the value of g. **[0.5]**

(c) Among eligible refreshment kiosks, find the smallest allowed sponsored payoff g at which the Minimax Regret recommendation changes from C_4 to C_1 alone. Recalculate regret benchmarks using only eligible kiosks. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $102 \leq g \leq 170$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

(a) Initial recommendations **[1]**

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	C_3	Row maximum: 190
Maximin	C_1	Row minimum: 68
Minimax Regret	C_4	Maximum regret: 42
EMV/Bayes	C_4	EMV: 85.6

All values are in hundred USD. Regret benchmarks use all four original kiosks, with state bests $(190, 68)$; no eligibility restriction applies in (a). Only EMV uses the probabilities; the other criteria compare upside, downside, or missed profit.



(b) Sponsored payoff and eligibility **[0.5]**

The updated payoff table is:

Alternative	High	Low
$P(S_j)$	0.35	0.65
Portable (C_1)	g	68
Full-service (C_2)	162	24
Deluxe (C_3)	190	-12
Fixed (C_4)	148	52

The updated decision statistics, before applying eligibility restrictions, are:

Alt.	Max	Min	Regret	EMV
C_1	g	68	$190 - g$	$0.35g + 44.2$
C_2	162	24	44	72.3
C_3	190	-12	80	58.7
C_4	148	52	42	85.6

An alternative is eligible only if both payoffs are at least 44. For integers $102 \leq g \leq 170$:

Alternative	Eligibility condition	Eligible values
C_1	$g \geq 44,\ 68 \geq 44$	$102 \leq g \leq 170$
C_2	$24 < 44$	None
C_3	$-12 < 44$	None
C_4	$148 \geq 44,\ 52 \geq 44$	$102 \leq g \leq 170$

The eligible set is always $\{C_1, C_4\}$.

(c) Minimax Regret change under sponsored payoff g [1.5]

Recalculate regrets using only these alternatives, with state benchmarks

$$b_H = \max\{g, 148\}, \qquad b_L = 68.$$

Their maximum regrets are

$$R(C_1) = \max\{148 - g, 0\},$$
$$R(C_4) = \max\{g - 148, 16\}.$$

For $102 \leq g \leq 148$, these become

$$R(C_1) = 148 - g, \qquad R(C_4) = 16.$$

For C_1 to be the sole Minimax Regret choice:

$$148 - g < 16 \quad \Longleftrightarrow \quad g > 132.$$

At $g = 132$, both alternatives have maximum regret 16. Thus, C_4 is the sole Minimax Regret choice for $102 \leq g \leq 131$; at $g = 132$, the recommendation becomes the tie $\{C_1, C_4\}$; and C_1 is the sole choice for $133 \leq g \leq 170$. For $g > 148$, C_1 has zero regret and C_4 has regret at least 16, so no further change occurs. The tie first changes the set of recommended kiosks, but a strict change to C_1 alone requires $g > 132$. The integer domain contains a smallest value satisfying this condition. Therefore, the smallest allowed integer giving a strict change is

$$g = 133\ .$$

At this value, the eligible payoff table is:

Alternative	High	Low
C_1	133	68
C_4	148	52

The corresponding maximum regrets are

$$R(C_1) = 15, \qquad R(C_4) = 16.$$

Thus, C_1 wins uniquely with $g = 133$, corresponding to \$13,300 in large-audience profit.

3 Garden Centre Stock [C9-2]

A garden centre selects a seedling stock plan D_1–D_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
D_1	159	71	66	134.8
D_2	181	49	44	150.2
D_3	137	93	88	121.6
D_4	203	27	66	150.2
D_5	115	71	110	99.6
D_6	225	5	88	150.2

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

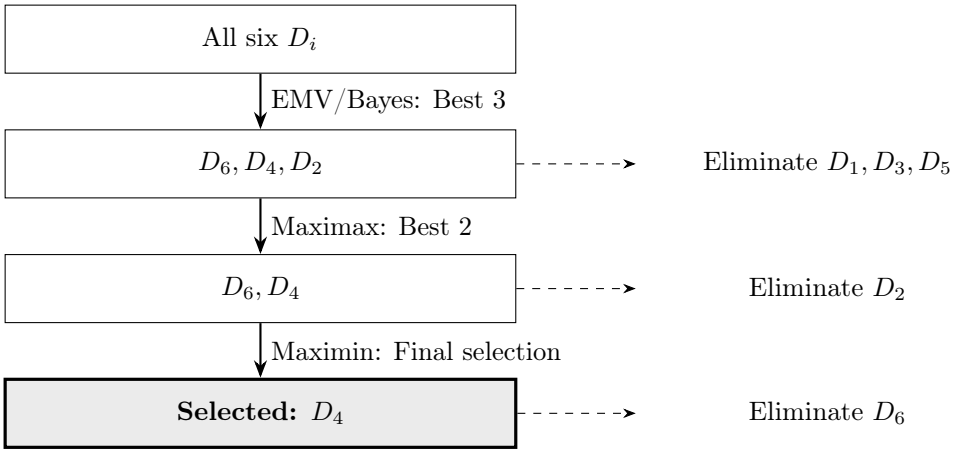
Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$D_6, D_4, D_2, D_1, D_3, D_5$	—
1	EMV/Bayes: Best 3	$D_6(150.2)$, $D_4(150.2)$, $D_2(150.2)$	$D_1(134.8)$, $D_3(121.6)$, $D_5(99.6)$
2	Maximax: Best 2	$D_6(225)$, $D_4(203)$	$D_2(181)$
3	Maximin: Final selection	**Selected:** $D_4(27)$	$D_6(5)$

The highest EMV, 150.2, is shared by D_6, D_4, D_2; all three fill the Best 3 places. Maximax retains D_6, D_4 with maxima 225 and 203, eliminating D_2 with 181. Maximin selects D_4 because $27 > 5$. There are no cutoff or final ties beyond the shared first-stage EMV.



Filtering tree and interpretation



The garden centre selects D_4: it survives the EMV and Maximax filters, then its minimum profit of 27 hundred USD exceeds D_6's 5.

4 Climbing Gym Expansion [C9-2]

A climbing gym selects an expansion plan F_1–F_6. Net profits are in thousand USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
F_1	178	52	45	130.3
F_2	142	88	81	122.2
F_3	160	70	63	138.4
F_4	223	7	81	116.8
F_5	124	79	99	106
F_6	196	34	54	123.1

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

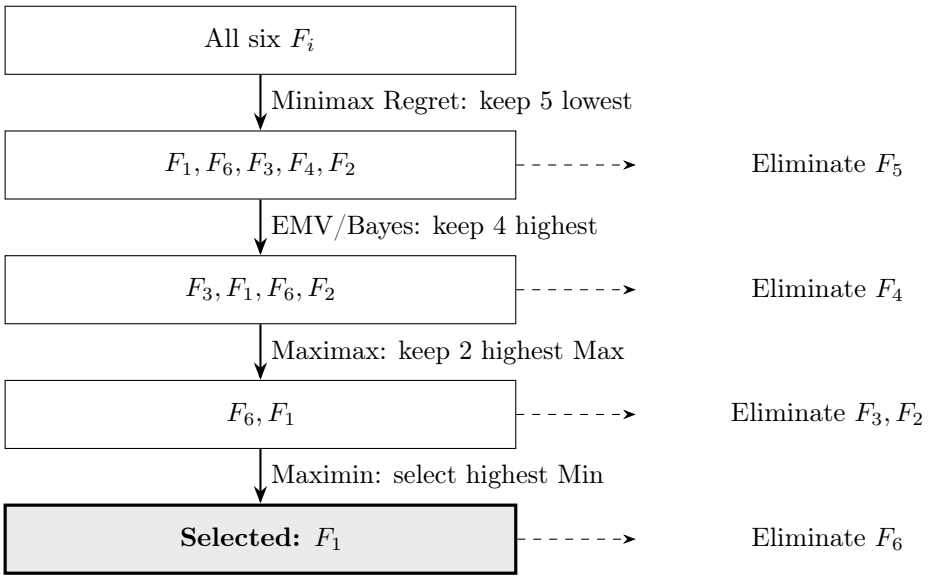
Illustrate each filtering stage, showing survivors and eliminations.

Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$F_4, F_6, F_1, F_3, F_2, F_5$	—
1	Minimax Regret: keep 5 lowest	$F_1(45)$, $F_6(54)$, $F_3(63)$, $F_4(81)$, $F_2(81)$	$F_5(99)$
2	EMV/Bayes: keep 4 highest	$F_3(138.4)$, $F_1(130.3)$, $F_6(123.1)$, $F_2(122.2)$	$F_4(116.8)$
3	Maximax: keep 2 highest Max	$F_6(196)$, $F_1(178)$	$F_3(160)$, $F_2(142)$
4	Maximin: select highest Min	**Selected:** $F_1(52)$	$F_6(34)$

At the first cutoff, F_4 and F_2 both have regret 81, and both survive. Later cutoffs have no ties.

Filtering tree and interpretation



The climbing gym selects F_1: after the regret, EMV, and upside filters, its minimum profit of 52 thousand USD exceeds F_6's 34.

5 Art Materials Orders [C9-2]

An art materials distributor selects an order plan H_1–H_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
H_1	178	106	108	151.6
H_2	202	82	84	170.8
H_3	250	34	72	152.8
H_4	154	94	132	130
H_5	286	-2	108	144.4
H_6	226	58	60	162.4

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

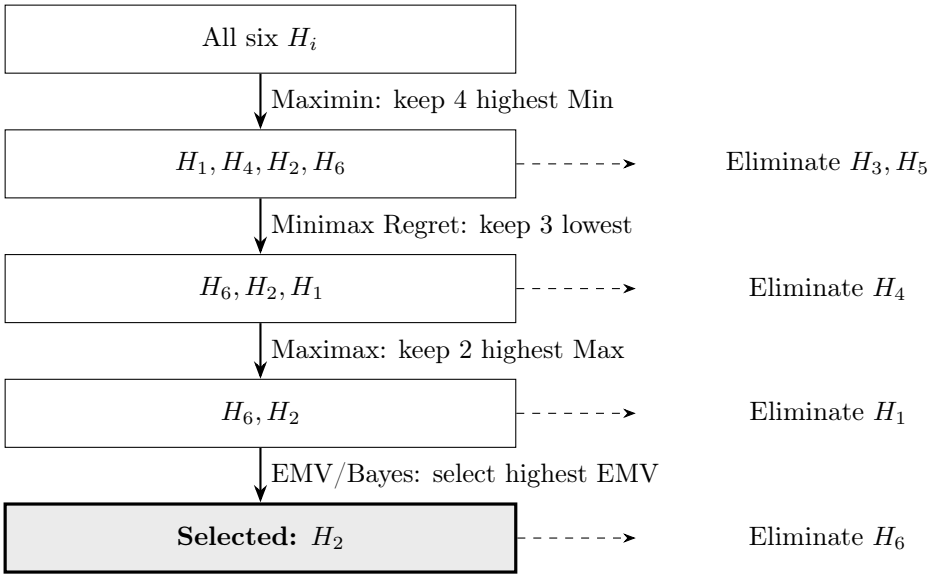
Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$H_5, H_3, H_6, H_2, H_1, H_4$	—
1	Maximin: keep 4 highest Min	$H_1(106)$, $H_4(94)$, $H_2(82)$, $H_6(58)$	$H_3(34)$, $H_5(-2)$
2	Minimax Regret: keep 3 lowest	$H_6(60)$, $H_2(84)$, $H_1(108)$	$H_4(132)$
3	Maximax: keep 2 highest Max	$H_6(226)$, $H_2(202)$	$H_1(178)$
4	EMV/Bayes: select highest EMV	**Selected:** $H_2(170.8)$	$H_6(162.4)$

The cutoffs are Min 58, maximum regret 108, and Max 202; the final EMV comparison is 170.8 versus 162.4. There are no cutoff or final ties.



Filtering tree and interpretation



The art materials distributor selects H_2: after the downside, regret, and upside filters, its EMV of 170.8 hundred USD exceeds H_6's 162.4.

C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.

C10 example solution.

C5 catch Up

Camera Rental Plans (Question 1). Apply Minimax Regret to the original payoff table.

A_1 Essential, A_2 Shared, A_3 Extensive, A_4 Modular.
S_1 High, S_2 Normal, S_3 Low.
Profits and regrets are in thousands of USD.
State bests are $\boldsymbol{b} = (250, 160, 65)$.
Regret is $r_{ij} = b_j - P_{ay}(A_i, S_j)$.

Regret	S_1	S_2	S_3	Max.
A_1	$250-110$ $=140$	$160-95$ $=65$	$65-65$ $=0$	140
A_2	$250-150$ $=100$	$160-160$ $=0$	$65-35$ $=30$	100
A_3	$250-250$ $=0$	$160-52$ $=108$	$65-(-35)$ $=100$	108
A_4	$250-180$ $=70$	$160-135$ $=25$	$65-(-15)$ $=80$	80

$$\min\{140, 100, 108, 80\} = 80 \implies A_4 \text{ (Modular)} \quad \text{Maximum regret } 80$$

6 Summarized Solutions [C8–C9]

6.1 Camera Rental Plans C8

Values in thousands of USD.

(a) Original payoffs

Criterion	Choice	Value
Maximax	A_3	250
Maximin	A_1	65
Minimax Regret	A_4	80
EMV/Bayes	A_2	132

(b) Eligibility

Alternative	Eligible range of s
A_1	$0 \le s \le 95$
A_2	$0 \le s \le 95$
A_3	$35 \le s \le 95$
A_4	$15 \le s \le 95$

(c) Maximin

- Compare eligible minima $m_1 = 65$ and $m_4 = s - 15$. Both $m_2 = 35$ and $m_3 \le 52$ are below m_1.
- A_1 and A_4 tie at $s = 80$ with minimum 65.
- First sole win for A_4 at $s = 81$, with minimum 66.

6.2 Outdoor Cinema Refreshments C8

Values in hundreds of USD.

(a) Original payoffs

Criterion	Choice	Value
Maximax	C_3	190
Maximin	C_1	68
Minimax Regret	C_4	42
EMV/Bayes	C_4	85.6

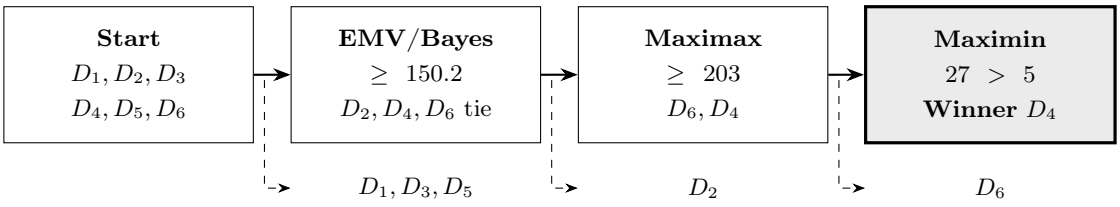
(b) Eligibility

Alternative	Eligible range of g
C_1	$102 \le g \le 170$
C_2	None
C_3	None
C_4	$102 \le g \le 170$

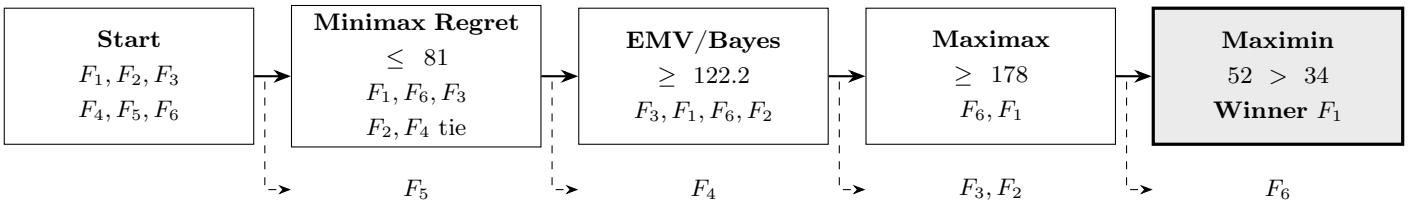
(c) Minimax Regret

- Eligible state bests (high, low) from C_1, C_4 are $(\max(g, 148), 68)$.
- Maximum regrets are $R_1 = \max(148 - g, 0)$ and $R_4 = \max(g - 148, 16)$.
- C_1 and C_4 tie at $g = 132$ with regret 16.
- First sole win for C_1 at $g = 133$, where $15 < 16$.

6.3 Garden Centre Stock C9



6.4 Climbing Gym Expansion C9



6.5 Art Materials Orders C9

